

year, including product sales, services, research projects, and prototype development.

The startup offers a range of advanced counter-drone solutions and unmanned aerial vehicles (UAVs), including handheld neutralisers, continuous operational omnidirectional neutralisers, compact UAVs, and net drones for capturing unauthorised drones.

Megh Robotics' 4-channel handheld neutraliser offsets unauthorised UAVs with a precision 45-degree angular frequency wave and disrupts the drone's remote controller, camera feed, and GPS signals. This device can operate within a one-kilometre range and has a battery life that lasts 75 minutes; it weighs 3.5kg.

For comprehensive coverage, the startup has introduced an 8-channel omni-directional neutraliser, which can handle globally known commercial UAV operational bands, operating 24/7 as a static system, portable solution, or vehicle-mounted unit. The device has a power output of 180W.

Chakraborty explains that the counter-drone system generates a power output of 20W per channel, necessitating a multilayer PCB that can decrease local heat without increasing the weight of the whole machine. The most used multilayer PCBs have plural layers such as 4, 8, or 16, but Megh Robotics has chosen to use a 3-layer PCB board, which is pretty uncommon but highly cost-effective. The tri-layer PCB decreases the component welding points, reducing the failure rate, and introduces a warm grounding place to lower local heat. It reduces the spacing of components, minimises the signal transmission path, and enhances wiring density, ultimately



4-channel handheld neutraliser



8-channel omnidirectional neutraliser



Sobhan Chakraborty, Founder

reducing the circuit's signal distortion with the shielding layer's help.

The startup has outsourced the manufacturing of the high-end PCBs of the omnidirectional neutraliser to a Bengaluru-based startup. At the same time, it manufactures the PCBs for the 4-channel handheld neutraliser through rapid prototyping at IIT Patna Incubation Centre, which has a PCB manufacturing lab and a clean room fabrication facility. The antennas are being sourced from a New Delhi-based startup.

With a limited range and time, the startup currently serves a target segment comprising state government departments, such as the police and government security forces, with the pricing range starting from ₹800,000. "We are creating a mid-segment in India by offering

an effective pricing without compromising product quality," says Chakraborty.

Identifying a gap in the market due to the limited knowledge surrounding the niche field of counter-drone systems, Chakraborty says that most existing manufacturers end up importing the entire product or key components without testing their reliability. When their devices fail to show promised results during trials or

after deployment, the companies approach Megh, for assembling, repairing, and maintaining their existing solutions. "Initially, the companies who were our competitors and supplied counter-drone systems to the government departments have now become collaborators," explains Chakraborty.

The startup has also worked on other products within the counter-drones category to expand its product portfolio. Megh has developed a customisable net drone that captures targeted drones with a string net to avoid collateral damage once the unauthorised drones are neutralised. This drone can fly for 40 minutes and has an integrated camera. Apart from that, the company has launched AI-based dual cameras equipped with a thermal imaging sensor with 30x optical zoom and a shell launcher system for advanced security applications.

"We are now focusing on the technology part of the design to improve the fly-time and range and provide the best possible technology solutions to our nation at a cost-effective price so that every department can access our products for securing the airspace. We intend to proceed further on a channel partnership or case-to-case basis across India," he concludes.

—Yashasvini Razdan